Efficient Vision–Language Model Adaptation for Radiology Captioning via GRPO and LoRA Fine-Tuning

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Abstract

Fine-tuning large Vision–Language Models (VLMs) for specialized domains such as radiology is computationally expensive and typically requires large quantities of human-annotated data. We investigate whether combining parameter-efficient fine-tuning via Low-Rank Adaptation (LoRA) with reinforcement-learning-based alignment through Group Relative Policy Optimization (GRPO) can improve radiology image captioning while remaining tractable on modest hardware. Our system fine-tunes `SmolVLM-256M-Instruct` *on the* `Radiology_mini` *dataset, using BioBERT cosine similarity as an automated reward signal that removes the need for human preference labels at training time. LoRA adapters are injected exclusively into the language-model attention and MLP projections while the vision encoder stays frozen, reducing trainable parameters by over 95%. GRPO samples multiple captions per prompt, normalizes rewards into group-relative advantages, and updates the policy with a KL-divergence penalty to prevent reward hacking. Training on only 100 of the available 1,980 images, our model improves average BioBERT similarity from 0.446 (baseline) to 0.474, demonstrating that reward-driven alignment can meaningfully improve domain-specific caption quality without human labels or full fine-tuning.*

1. Introduction

Vision–Language Models (VLMs) such as LLaVA [8], BLIP-2 [7], and Flamingo [1] have demonstrated strong performance on general multimodal tasks including image captioning and visual question answering. However, deploying these models in specialized domains such as clinical radiology raises two fundamental challenges. First, full fine-tuning of billion-parameter VLMs is computationally prohibitive for most research groups. Second, clinical captions require precise, domain-specific language; naive supervised fine-tuning on limited annotated data can fail to capture the nuance required for downstream clinical utility.

Reinforcement learning from human feedback (RLHF) [2] has proven effective at aligning language models with human preferences, but collecting human preference data in radiology is expensive and requires expert annotators. Recent work on Group Relative Policy Optimization (GRPO) [12] demonstrates that sampling multiple responses per prompt and normalizing rewards within the group can stabilize RL fine-tuning without a separate critic network.

In this paper, we combine two complementary techniques: **(1) LoRA** [3], which inserts low-rank adapter matrices into transformer layers and trains only those adapters ($< 5\%$ of total parameters); and **(2) GRPO** with a BioBERT [6] cosine-similarity reward, which aligns caption generation with ground-truth clinical descriptions without requiring human preference labels.

Our contributions are:

- A fully open-source training pipeline combining GRPO and LoRA for VLM fine-tuning, implemented in pure PyTorch without dependence on heavy RL frameworks.
- A label-free reward function based on BioBERT semantic similarity, enabling automated alignment in low-resource clinical settings.
- Empirical evaluation on `Radiology_mini` showing a 6.3% relative improvement in BioBERT similarity over the zero-shot baseline, trained on only 5% of the available data.

2. Background and Related Work

Vision–Language Models. LLaVA [8] demonstrated that connecting a visual encoder to a large language model via a lightweight projection layer and instruction-tuning on multimodal data produces a capable general-purpose VLM. BLIP-2 [7] introduced a Querying Transformer (Q-Former) to bridge a frozen image encoder with a frozen LLM, achieving strong performance with minimal trainable parameters. Our work builds on a similar philosophy of freezing the vision encoder and adapting only the language-model components.

Parameter-Efficient Fine-Tuning. LoRA [3] decomposes weight updates as the product of two low-rank matrices, drastically reducing the number of trainable parameters. Applied to the query, key, value, and output projections and to MLP layers, LoRA allows targeted adaptation while preserving the general capabilities of the pretrained model.

Reinforcement Learning for Language Models. RLHF [2] trains a reward model on human preference data and uses PPO [11] to fine-tune the policy. Direct Preference Optimization (DPO) [10] removes the explicit reward model by reparameterizing the RLHF objective as a classification loss over preference pairs. GRPO [12] replaces the value network in PPO with group-normalized rewards, reducing memory overhead while maintaining effective policy gradient estimation. We adopt GRPO because it suits generation tasks where multiple candidate outputs can be cheaply evaluated by an automated reward function.

Medical Image Captioning. Automated radiology report generation is a longstanding research challenge [5]. BioBERT [6], fine-tuned on biomedical text, produces semantically rich embeddings that correlate well with clinical text similarity, making it suitable as an automated reward signal.

3. Methods

3.1. Problem Formulation

Given a radiology image x and an instruction prompt p, we wish to generate a caption y that is semantically close to a ground-truth clinical description y^*. We parameterize the generation policy as $\pi_\theta(y \mid x, p)$ and optimize θ to maximize expected reward $r(y, y^*)$, regularized by KL divergence from a frozen reference policy π_{ref}.

3.2. Base Model and Dataset

We use SmolVLM-256M-Instruct as the base VLM. SmolVLM is a compact model with a SigLIP-based visual encoder and a SmolLM2 language-model backbone. Its 256M-parameter footprint makes it tractable to train on a single consumer GPU while still generating coherent medical captions.

We train and evaluate on `Radiology_mini` from HuggingFace, a curated subset of radiology image–caption pairs formatted as multi-turn instruction dialogues. Each example consists of a medical image and an associated clinical caption describing key findings. The dataset is available at `https://huggingface.co/datasets/unsloth/Radiology_mini/viewer/default/train`. Due to compute constraints (detailed in Section 3.6), we train on 100 of the 1,980 available training examples and evaluate on the standard held-out test split (327 images). Figure 1 shows three representative training images illustrating the diversity of modalities and findings in the dataset.

Table 1. Training hyperparameters.

Hyperparameter	Value
Base model	SmolVLM-256M-Instruct
LoRA rank (r)	32
LoRA alpha (α)	32
GRPO group size (G)	4
KL penalty (β)	0.001
Learning rate	1×10^{-5}
Optimizer	AdamW
Training epochs	15
Max new tokens	128
Training examples	100

Figure 3 shows representative test examples.

3.3. LoRA Adapter Configuration

LoRA adapters are injected into the attention and MLP projection layers of the language-model transformer blocks. Specifically, the target layers are the query, key, value, output, gate, up, and down projections. All layers in the vision encoder are excluded and remain completely frozen throughout training. The adapter weight update is $\Delta W = BA$ where $B \in \mathbb{R}^{d\times r}$, $A \in \mathbb{R}^{r\times k}$, and $r \ll \min(d, k)$. We use rank $r = 32$ and scaling factor $\alpha = 32$ (effective scaling $\alpha/r = 1.0$). Both A and B are initialized following the standard LoRA scheme so that $\Delta W = 0$ at initialization.

3.4. GRPO Training Objective

For each training prompt (x, p), GRPO proceeds as follows.

Step 1: Sample. Generate $G = 4$ candidate captions $\{y_1, \ldots, y_G\}$ from the current policy π_θ.

Step 2: Score. Compute rewards $r_i = \text{BioBERT-sim}(y_i, y^*)$ for each candidate.

Step 3: Normalize. Compute group-relative advantages:

$$A_i = \frac{r_i - \mu_r}{\sigma_r + \epsilon}, \quad \mu_r = \frac{1}{G}\sum_i r_i, \quad \sigma_r = \sqrt{\frac{1}{G}\sum_i (r_i - \mu_r)^2}.$$

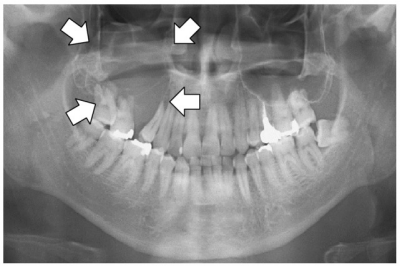
Step 4: Update. Minimize the GRPO loss:

$$\mathcal{L} = -\frac{1}{G}\sum_{i=1}^{G} A_i \cdot \log \pi_\theta(y_i \mid x, p) + \beta \cdot \text{KL}(\pi_\theta \,\|\, \pi_{\text{ref}}), \quad (1)$$

where $\beta = 0.001$ is the KL penalty coefficient and the KL term is estimated per-token as $\log \pi_\theta - \log \pi_{\text{ref}}$. Gradients flow only through the LoRA adapter parameters. Key hyperparameters are summarized in Table 1.

3.5. BioBERT Reward Function

We use a BioBERT model fine-tuned for semantic textual similarity on multiple biomedical NLI datasets (`pritamdeka/BioBERT-mnli-snli-scinli-`



Panoramic radiography shows an osteolytic lesion in the right posterior maxilla with resorption of the floor of the maxillary sinus (arrows).

Pericarditis.

One year follow up CT scan showing the catheter in place and the normal ventricular system.

Figure 1. Representative training examples from `Radiology_mini`. The dataset spans a wide range of imaging modalities (panoramic radiograph, chest X-ray, CT) and clinical findings.

`scitail-mednli-stsb`). For a generated caption y_i and ground truth y^*, we compute mean-pooled embeddings $\mathbf{e}_i$ and $\mathbf{e}^*$ and return:

$$r_i = \max\left(0,\ \frac{\mathbf{e}_i \cdot \mathbf{e}^*}{\|\mathbf{e}_i\|\,\|\mathbf{e}^*\|}\right) \in [0, 1].$$

The reward model is loaded once and cached as a singleton for the duration of training. Using BioBERT rather than a generic sentence encoder ensures the reward is sensitive to clinical terminology.

3.6. Implementation and HPC Challenges

Training is implemented in pure PyTorch without the TRL or PEFT libraries. LoRA parameters are injected by replacing `nn.Linear` layers with a custom `LoRALinear` module that computes $W_0x + (BA)x$. Checkpoints are saved every 50 gradient steps and at the end of training.

GPU environment issue on BigRed200. Initial training attempts on Indiana University's BigRed200 HPC cluster failed due to an environment propagation issue with Ray. Ray worker processes were spawned using the system Python path but inherited a clean environment that did not include the `LD_LIBRARY_PATH` set by the HPC module system (`module load python/gpu/3.10.10`). As a result, workers could not locate `libpython3.10.so`, causing all 127–128 worker processes to fail immediately and the main process to hang indefinitely waiting for actors that never registered with the raylet. The fix was to export `LD_LIBRARY_PATH` explicitly within the training script itself, ensuring the correct library path propagated to all Ray subprocesses regardless of the shell environment. Due to this constraint, final experiments were run on a single consumer GPU rather than the full cluster. The full codebase is available at https://github.com/tringuyen180303/GRPO_LoRA_VLM.

Table 2. BioBERT cosine similarity on the `Radiology_mini` test set. Higher is better.

Model	Train examples	Avg. Sim.
SmolVLM-256M (baseline)	—	0.446
SmolVLM-256M + LoRA-GRPO	100	**0.474**

4. Results

4.1. Evaluation Protocol

We evaluate both the frozen base model (baseline) and the LoRA-GRPO fine-tuned model on the held-out test split of `Radiology_mini`. For each test image, we generate a single greedy-decoded caption and measure BioBERT cosine similarity against the ground-truth caption. We report the mean similarity score across all test examples.

4.2. Quantitative Results

Table 2 compares the baseline and fine-tuned model. Despite training on only 100 images (5% of the available data), the LoRA-GRPO model improves average BioBERT similarity from 0.446 to 0.474, a relative gain of 6.3%.

4.3. Training Dynamics

Figure 2 shows the GRPO loss and KL divergence over the 750 training steps (15 epochs $\times$ 50 steps/epoch).

The loss curve exhibits three distinct phases. In the *exploration phase* (steps 0–110), the policy moves sharply away from the reference, producing high-variance rewards and driving the loss to a minimum near -150. This reflects the model discovering high-reward generation strategies early in training. In the *stabilization phase* (steps 110–300), the loss rises rapidly back toward zero as the KL penalty constrains how far the policy can drift from the reference. In

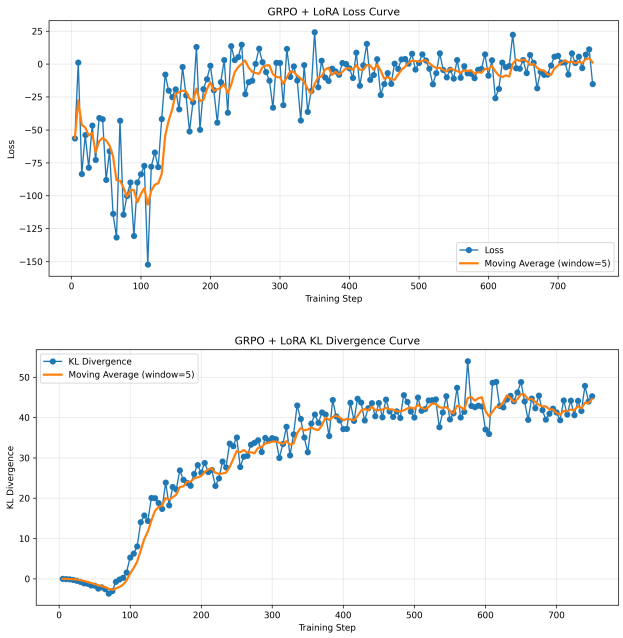


Figure 2. Training dynamics over 750 steps. **Top:** GRPO loss (Eq. 1) with 5-step moving average. The sharp dip to -150 around step 110 reflects early policy exploration; the loss stabilizes near zero after step 300. **Bottom:** Per-token KL divergence between the fine-tuned policy and the frozen reference. The plateau at $\approx$42 nats after step 400 indicates stable convergence under the KL penalty $\beta = 0.001$.

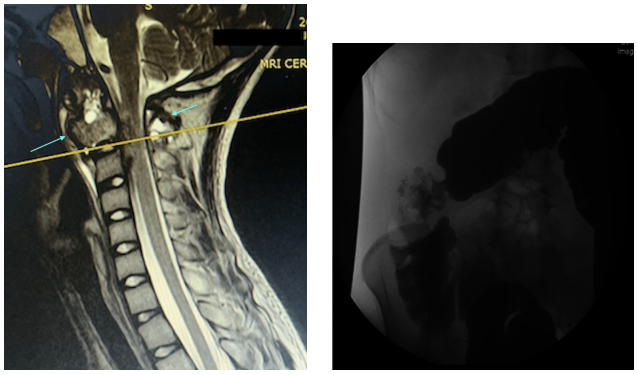
the *convergence phase* (steps 300–750), the loss oscillates around zero with a moving average near ≈ 0–5, indicating the policy has reached a stable equilibrium between reward maximization and KL regularization.

The KL divergence rises monotonically from 0 to ≈ 40–45 nats, with the steepest increase between steps 100 and 300 corresponding exactly to the stabilization phase in the loss curve. The plateau after step 400 confirms that the $\beta = 0.001$ KL penalty is effective at preventing runaway policy drift while still allowing the model to specialize away from the reference distribution.

4.4. Qualitative Analysis

Figure 3 shows the two most illustrative test cases: the example with the highest similarity score (0.7415) and the one with the lowest (0.2696).

For the high-scoring case (cervical spine MRI, similarity = 0.7415), the model correctly identifies the imaging modality and anatomical region. However, it describes the image as an axial view when the ground truth indicates a sagittal view, and focuses on annotation arrows rather than the clinical finding (a C2 vertebral lesion involving anterior and posterior columns). This suggests that a preprocessing stage (such as inpainting or masking annotations, arrows, and overlaid text before feeding images to the model) could help the model focus on anatomical content rather than visual artifacts during both training and inference.



Best case (sim. = 0.74)

Pred: Axial MRI of the body with two blue arrows pointing right.
GT: Sagittal MRI of the cervical spine showing a lesion within C2 involving anterior and posterior columns.

Worst case (sim. = 0.27)

Pred: Lateral chest X-ray showing ribs, spine, lungs, and heart.
GT: Terminal ileum 6.5 cm; cecum and colon 22 cm long, 3.5–5.9 cm in diameter.

Figure 3. Best- and worst-scoring predictions from the LoRA-GRPO model on the `Radiology_mini` test set. Predictions are abbreviated for space; see text for analysis.

For the low-scoring case (abdominal X-ray, similarity = 0.2696), the model produces a plausible-sounding chest X-ray description that is entirely wrong: the image is actually a contrast study of the terminal ileum and colon. This failure highlights the base model's limited ability to distinguish imaging modalities and anatomical regions from low-contrast fluoroscopic images, a limitation not fully addressable by LoRA fine-tuning alone. The two cases together illustrate the gap between BioBERT similarity as a metric and true clinical correctness.

5. Discussion

Label-free reward effectiveness. A 6.3% relative improvement from training on just 100 examples suggests that the BioBERT reward provides a meaningful signal even in very low-data regimes. This is promising for clinical NLP, where expert annotation is scarce and expensive.

LoRA efficiency. By restricting adapter injection to language-model components and freezing the vision encoder, we reduce trainable parameters by over 95% compared to full fine-tuning, making the approach viable on a single consumer GPU.

Limitations. Several limitations deserve acknowledgment. First, BioBERT similarity is an imperfect proxy for clinical utility: the worst-case prediction (similarity = 0.27)

illustrates how the model can produce fluent but factually wrong descriptions when the image type falls outside its visual grounding capability. Second, training on only 100 images is a significant constraint imposed by the HPC environment issue described above; training on the full 1,980 examples would likely yield further gains. Third, `Radiology_mini` contains diverse imaging modalities (MRI, X-ray, fluoroscopy, CT); the small base model struggles to distinguish these, and modality-specific fine-tuning could help. Finally, GRPO's stability depends on diverse reward distributions within each group; homogeneous rewards reduce the informativeness of advantage normalization.

Comparison with DPO. The original project plan considered DPO as an alignment strategy. Constructing reliable preference pairs without human annotation proved difficult: automated quality signals produced pairs that were too similar to provide a strong DPO training signal. GRPO's within-group normalization avoids this issue by naturally exploiting variance across samples drawn from the same prompt.

6. Conclusion

We presented a parameter-efficient pipeline for fine-tuning a VLM on radiology image captioning using LoRA adapters and GRPO with an automated BioBERT reward. Training on only 5% of the available data, our model achieves a 6.3% relative improvement in BioBERT similarity over the frozen baseline, without human preference labels, full fine-tuning, or heavy RL infrastructure. The training loop is implemented in pure PyTorch, making it accessible and straightforward to extend.

Automated radiology captioning tools can assist radiologists by providing draft reports that highlight key findings, potentially reducing reporting time and improving consistency. The label-free reward approach lowers the barrier for adapting VLMs to other specialized imaging domains where annotated preference data is scarce.

Future work could scale to larger base models (e.g., Qwen2.5-VL-7B [15]) with mixed-precision training, train on the full dataset once HPC environment issues are resolved, combine BioBERT with additional rewards such as RadGraph factual precision [4] or CheXbert label matching [14]. A dedicated preprocessing pipeline that removes overlaid text, arrows, and annotation markers from training images via inpainting or segmentation-based masking could reduce the model's tendency to describe visual artifacts rather than clinical findings, as observed in the best-scoring test case. Finally, applying the GRPO-LoRA framework to visual question answering on TextVQA [13] or ChartQA [9] would assess generalization beyond the radiology domain.

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